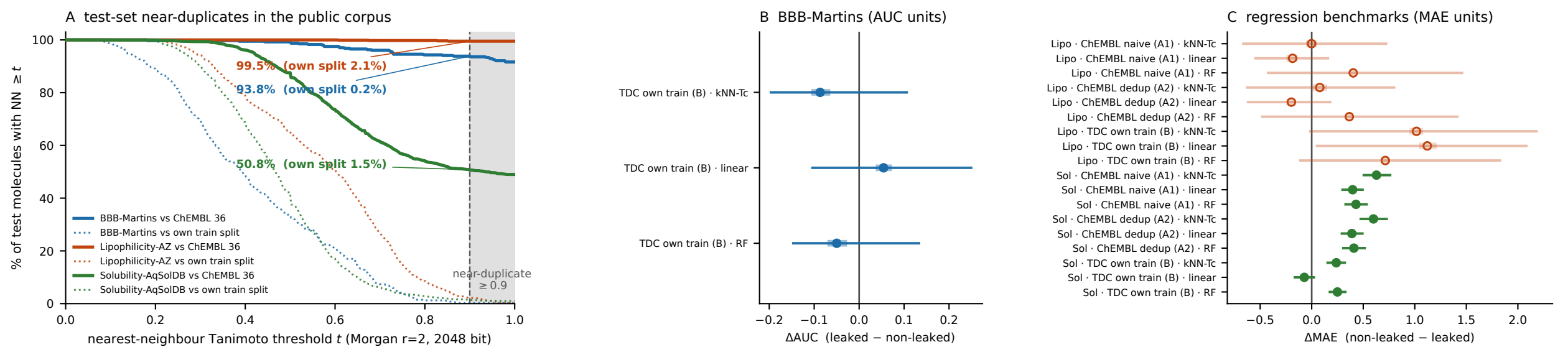
A test-set near-duplicates in the public corpus
% of test molecules with NN ≥ t
nearest-neighbour Tanimoto threshold t (Morgan r=2, 2048 bit)
99.5% (own split 2.1%)
93.8% (own split 0.2%)
50.8% (own split 1.5%)
near-duplicate ≥ 0.9
BBB-Martins vs ChEMBL 36
BBB-Martins vs own train split
Lipophilicity-AZ vs ChEMBL 36
Lipophilicity-AZ vs own train split
Solubility-AqSolDB vs ChEMBL 36
Solubility-AqSolDB vs own train split
B BBB-Martins (AUC units)
TDC own train (B) · kNN-Tc
TDC own train (B) · linear
TDC own train (B) · RF
ΔAUC (leaked − non-leaked)
C regression benchmarks (MAE units)
Lipo · ChEMBL naive (A1) · kNN-Tc
Lipo · ChEMBL naive (A1) · linear
Lipo · ChEMBL naive (A1) · RF
Lipo · ChEMBL dedup (A2) · kNN-Tc
Lipo · ChEMBL dedup (A2) · linear
Lipo · ChEMBL dedup (A2) · RF
Lipo · TDC own train (B) · kNN-Tc
Lipo · TDC own train (B) · linear
Lipo · TDC own train (B) · RF
Sol · ChEMBL naive (A1) · kNN-Tc
Sol · ChEMBL naive (A1) · linear
Sol · ChEMBL naive (A1) · RF
Sol · ChEMBL dedup (A2) · kNN-Tc
Sol · ChEMBL dedup (A2) · linear
Sol · ChEMBL dedup (A2) · RF
Sol · TDC own train (B) · kNN-Tc
Sol · TDC own train (B) · linear
Sol · TDC own train (B) · RF
ΔMAE (non-leaked − leaked)
thin bar: molecule-level 95% CI · thick bar: seed-level 95% CI
hollow marker / faded bar = fewer than 20 non-leaked test molecules